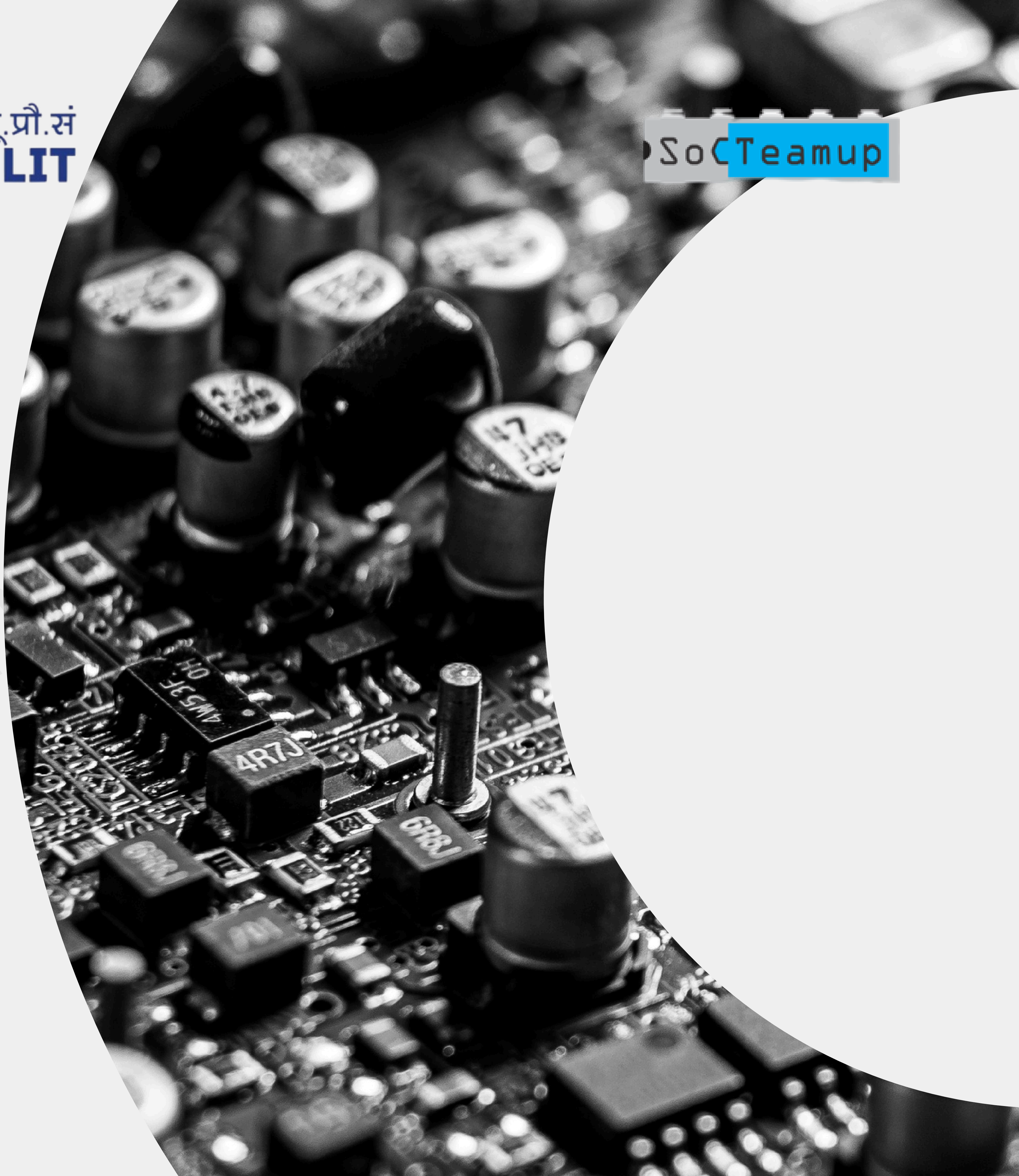
इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY

रा.इ.सू.प्रौ.सं
NIELIT

SoCTeamup

IP Design Methodology

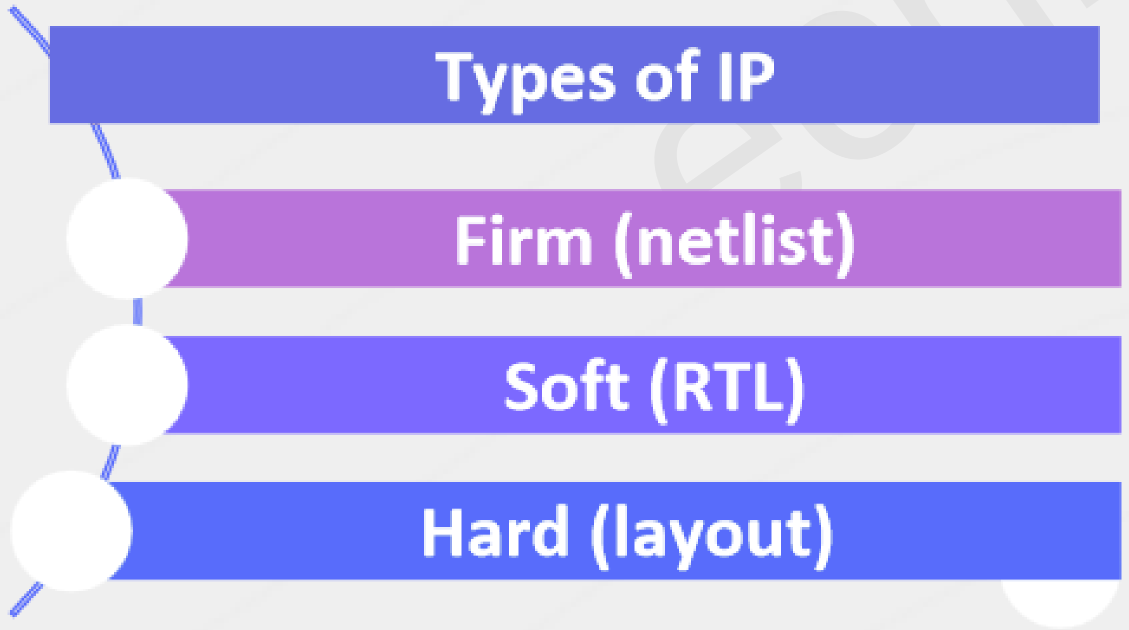


Outline

- What is IP in VLSI?
- IP Design Lifecycle: Overview
- Step 1: Specification
- Step 2: RTL Design
- Step 3: Verification
- Summary of Best Practices

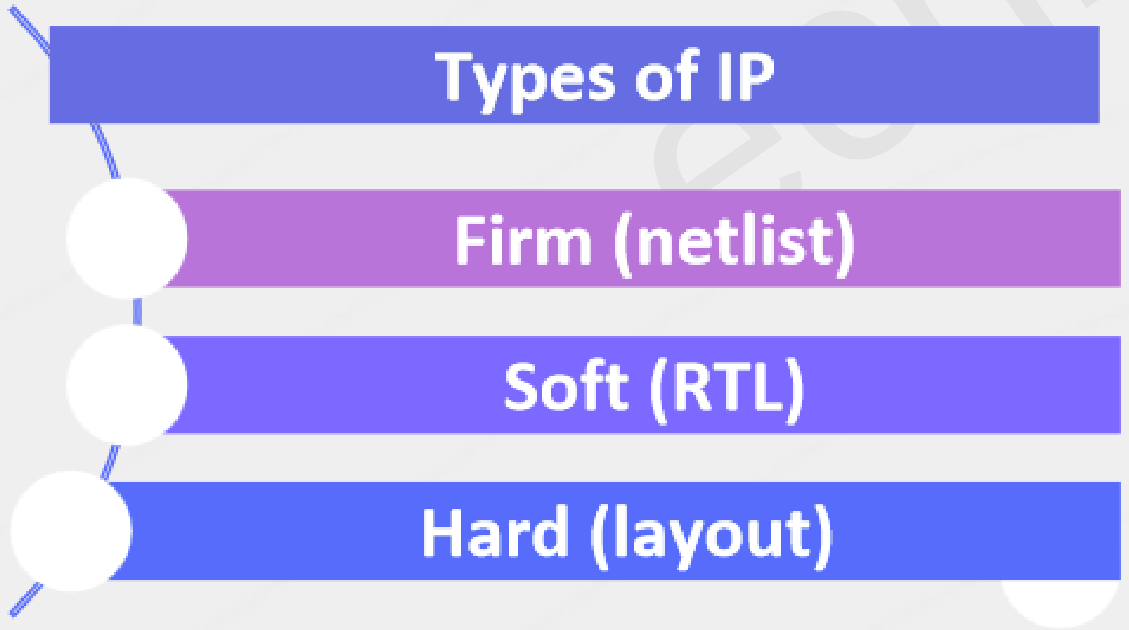
What is IP in VLSI?

- **IP : Intellectual Property (reusable logic block)**
  - Pre-designed, pre-verified functional block
  - Enables rapid SoC development and integration



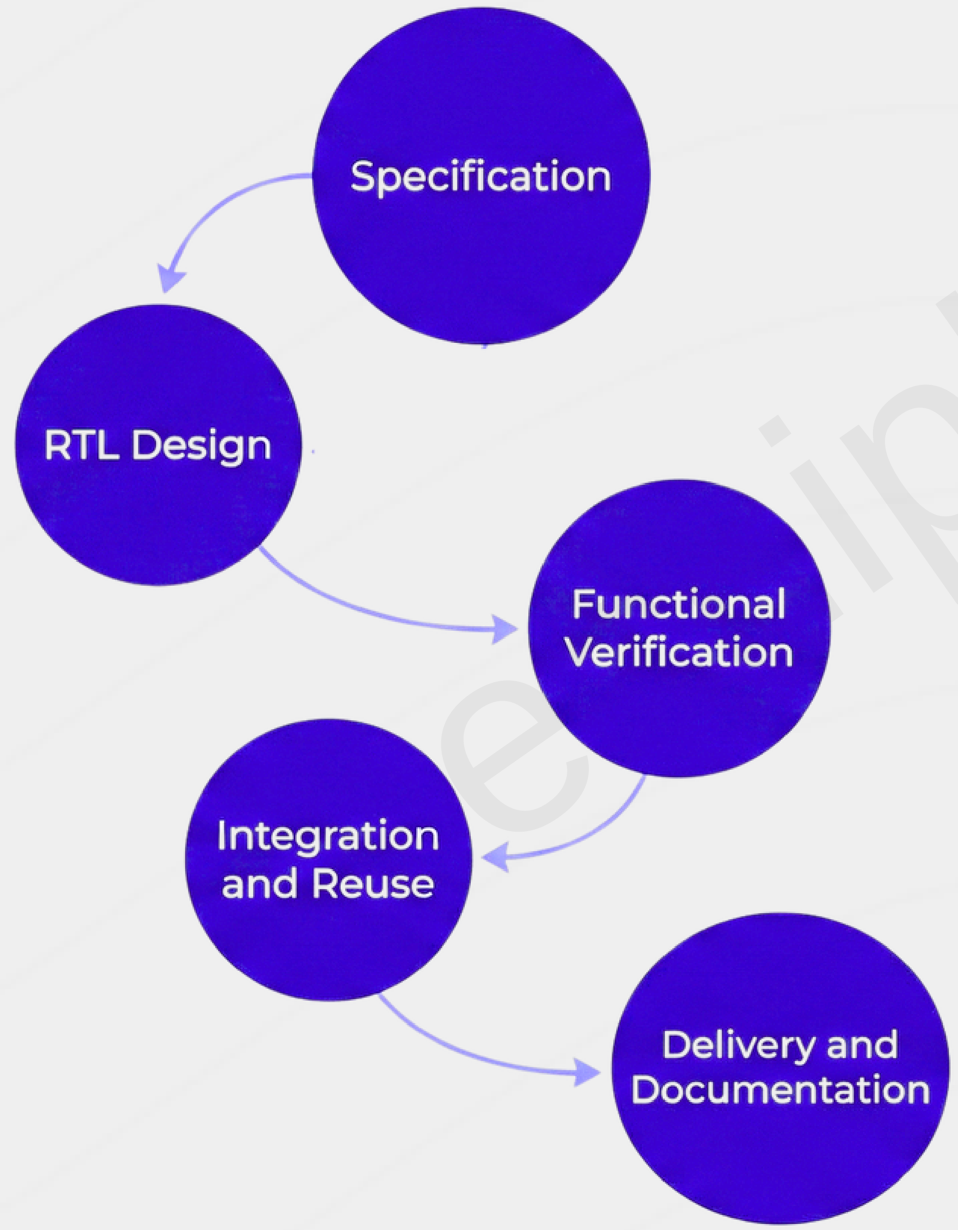
Examples:

- UART controller
- SPI master/slave
- RISC-V core



IP Design Lifecycle – Overview

Goal: Reusable, verified, modular logic for SoC or FPGA design



Step 1 :Specification

Defines:

Deliverables

Step 2 : RTL Design

Tasks:

Best Practices:

Parameterization & Reusability

- **Why Parameterize?**
  - Adapt same IP to multiple SoCs/FPGAs
  - Make IP configurable (e.g., DATA_WIDTH, DEPTH)

Parameterization & Reusability

- **Verilog Example:**
  - module fifo #(parameter DATA_WIDTH = 8, DEPTH = 16) (...);

इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY

रा.इ.सू.प्रौ.सं NIELIT

SoCTeamup

Parameterization & Reusability

- **Benefits:**
    - Scalability
    - Flexibility
    - Code reuse across projects

Step 3 : Verification

- **Verification Objectives:**
  - Match behavior to spec
  - Validate corner cases
  - Check register/interface compliance

इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY

रा.इ.सू.प्रौ.सं NIELIT

SoCTeamup

Step 3 : Verification

- **Testbench Tools:**
  - Icarus Verilog
  - Verilator
  - SystemVerilog/UVM (for complex IPs)

Step 3 : Verification

- **Outputs:**
  - Coverage reports
  - Sign-off checklist

Summary of Best Practices

- Start from a clear written specification
- Modular, clean, synthesizable RTL
- Parameterize for reuse
- Strong testbench with coverage tracking
- Maintain versioning and documentation
- **Tools Mentioned:** Git, Verilator

इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY

रा.इ.सू.प्रौ.सं NIELIT

SoCTeamup

Thank You